M2: Formal LIMO ISA Specification (Frozen)

1. Architectural State (Registers)

LIMO features a **32-register file** (R0–R31). This preserves the 5-bit register fields standard in RV32I, ensuring easy translation to Ripes for testing.

Number	Sesotho Name	Alias	Type	Purpose
R0	Lefela	Zero	SPR	Hard-wired to 0. Writes ignored.
R1	Aterese ea Khutlo	Ra	SPR	Return address.
R2	Sesupa Pokello	Sp	SPR	Stack pointer.
R3	Sesupa Kakaretso	Gp	SPR	Global pointer.
R4	Sesupa Mola	Tp	SPR	Thread pointer.
R5-R7	Nakoana 1-3	t0-t2	GPR	Temporaries.
R8-R9	Bolokiloeng 1-2	s0-s1	GPR	Saved registers.
R10-R17	Khang 1-8	a0-a7	GPR	Function arguments / returns.
R18-R27	Bolokiloeng 3-12	s2-s11	GPR	Saved registers.
R28-R31	Nakoana 4-7	t3-t6	GPR	Temporaries.
PC	Sesupa Taelo	pc	SPR	Program Counter.

2. Instruction Formats

LIMO uses fixed 32-bit instructions. The field widths match RV32I exactly (5-bit rd, rs1, rs2).

- **R-Type:** funct7 (7) | rs2 (5) | rs1 (5) | funct3 (3) | rd (5) | opcode (7)
- **I-Type:** imm[11:0] (12) | rs1 (5) | funct3 (3) | rd (5) | opcode (7)
- **S-Type:** imm[11:5] (7) | rs2 (5) | rs1 (5) | funct3 (3) | imm[4:0] (5) | opcode (7)
- **B-Type:** imm[12|10:5] (7) | rs2 (5) | rs1 (5) | funct3 (3) | imm[4:1|11] (5) | opcode (7)
- **J-Type:** imm[20|10:1|11|19:12] (20) | rd (5) | opcode (7)

3. Opcode and Function-Field Assignments (The Assembler Blueprint)

This is the exact mapping Setente needs for the assembler logic.

Format	Opcode	Mnemonic	funct3	funct7	RV32I Equivalent
R	0110011	kpn (add)	000	0000000	add
R	0110011	tls (sub)	000	0100000	sub
R	0110011	le (and)	111	0000000	and
R	0110011	kp (or)	110	0000000	or
R	0110011	kpe (xor)	100	0000000	xor
R	0110011	stts (sll)	001	0000000	sll
R	0110011	stto (srl)	101	0000000	srl

Format	Opcode	Mnemonic	funct3	funct7	RV32I Equivalent
I	0010011	kpnh (addi)	000	—	addi
I	0010011	leh (andi)	111	—	andi
I	0010011	kph (ori)	110	—	ori
I	0010011	kpeh (xori)	100	—	xori
I	0000011	tsl (lw)	010	—	lw
S	0100011	bl (sw)	010	—	sw
B	1100011	khl (beq)	000	—	beq
B	1100011	khsl (bne)	001	—	bne
B	1100011	khn (blt)	100	—	blt
B	1100011	khk (bge)	101	—	bge
I	1100111	tlkr (jalr)	000	—	jalr
J	1101111	tlk (jal)	—	—	jal
J	1101111	tl (j)	—	—	jal x0, offset

M2 Sprint Note / D1 Feedback Log
Role: ISA Architect
Status: ISA Frozen for M2.
D1 Feedback Applied:
During D1, the examiner noted that our ISA completely drops RV32I's immediate shift instructions (slli, srli).
Decision: We have retained the original M1 decision to *only* use register-based shifts (stts, stto). This simplifies the datapath's shifter logic (the ALU only needs to read rs2 for shifts, and we don't need extra multiplexing for shamt). We accept that software must load the shift amount into a register first.